

Media Summary - FUND-FOP-047

Golden Ratio , e , and Fine Structure Constant : Collapse Breathing Proportions

For general media and public understanding (approx. 275 words)

Nature’s Hidden Mathematical Harmony: The Unexpected Connection Between Fundamental Constants

Have you ever wondered why certain numbers seem to appear repeatedly throughout nature? Scientists have long been puzzled by mysterious constants like the golden ratio (found in seashells and flower petals) and the fine structure constant (which controls how atoms behave). A groundbreaking new study reveals these seemingly unrelated numbers may actually be deeply connected through an elegant mathematical relationship.

Researchers Haobo Ma and Wen Niu have discovered that four fundamental constants—the golden ratio (approximately 1.618), Euler’s number e (approximately 2.718), π (approximately 3.14159), and the fine structure constant (approximately $1/137$)—form what they call “collapse breathing proportions.” This means these constants maintain specific mathematical relationships that remain stable through various transformations, similar to how your lungs maintain a consistent rhythm while breathing.

“For decades, physicists have wondered why the fine structure constant has its particular value,” explains Ma. “Our research suggests it’s not arbitrary but is precisely determined by other fundamental constants through specific information processing principles.”

The researchers used a novel approach called Universe Ontology, which views reality as emerging from information processes rather than physical particles. Through this lens, they derived exact mathematical formulas connecting these constants and verified their findings through high-precision numerical simulations.

This discovery could have profound implications for our understanding of reality. Rather than seeing physical constants as arbitrary values that happened to allow our universe to exist, they may be inevitable consequences of the underlying mathematical structure of information itself.

“These relationships suggest a deeper unity in nature than previously recognized,” says Niu. “The constants that govern our universe appear to be intricately linked in a harmonious mathematical symphony.”

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